CSE 260: Digital Logic Design

Lab Report

Experiment Name: Universal Gates and Applications of Boolean algebra

Submitted By:

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Name of the Experiment : Universal Gates and Applications of Boolean algebra.

Objective :

- To investigate the rules of Boolean algebra.
- To gain experience working with practical circuits.
- To simplify a complex function using Boolean algebra.

Required Components and Equipments :

- AT-700 Portable Analog/ Digital Labratory
- IC-7400 (NAND)
- wires
- Breadboard

Experimental Setup :

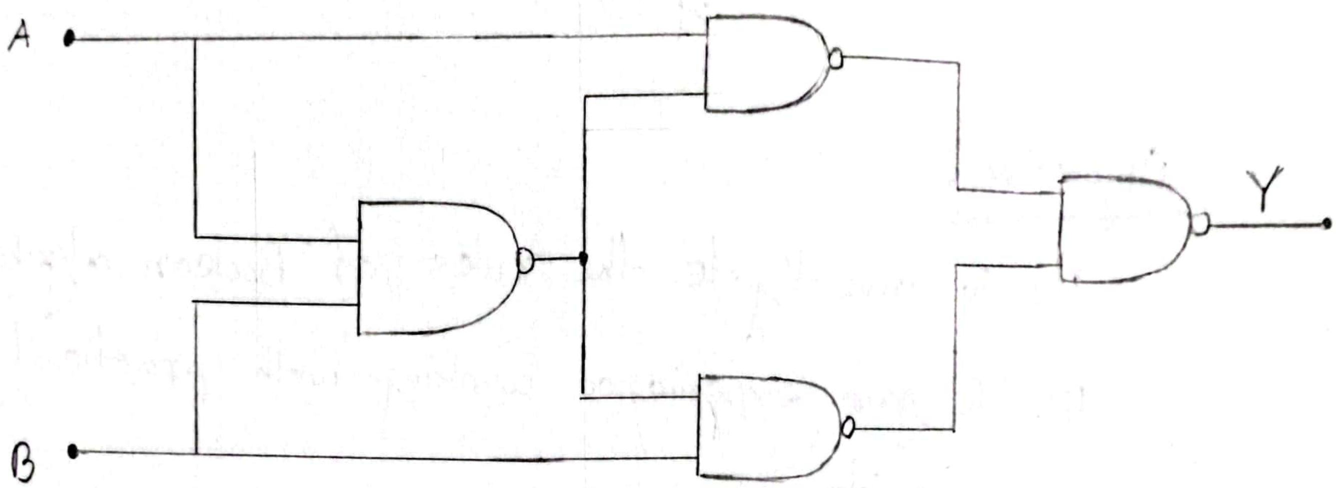

fig : Building Circuit using Universal Gates (NAND)

Truth Table

Input				Output
A	B	$\bar{A}$	$\bar{B}$	
0	0	1	1	0
1	1	0	0	0
1	0	0	1	1
0	1	1	0	1

Solving the equation:-

$$y = \overline{(\bar{A} + (AB)) \cdot (\bar{B} + (AB))}$$

$$\Rightarrow \overline{(\bar{A} + (AB))} + \overline{(\bar{B} + (AB))}$$

$$\Rightarrow \bar{\bar{A}} \cdot (\overline{AB}) + \bar{\bar{B}} \cdot (\overline{AB})$$

$$\Rightarrow A \cdot (\bar{A} + \bar{B}) + B \cdot (\bar{A} + \bar{B})$$

$$\Rightarrow (\bar{A} + \bar{B}) \cdot (A + B)$$

$$\Rightarrow A \cdot \bar{A} + \bar{A}B + A\bar{B} + B\bar{B}$$

$$y \Rightarrow \bar{A}B + A\bar{B}$$

10.10.22

$\therefore$ The truth table and the equation implies that the setup shown above acts like a XOR gate.

Discussion :

During the experiment, we made 4 NAND gates using 1 IC. Each of it has 2 inputs and after solving the circuit, we got the equation, for the output,

$$Y = \overline{\overline{(A \cdot (\overline{AB}))} \cdot \overline{(B(\overline{AB}))}}$$

Simplifying it, the final outcome is $\bar{A}B + A\bar{B} = A \oplus B$; which is a X-OR gate

For the gate, we connected the IC vertically on the e-f column of the breadboard. Inserting pin 14 with +5V and pin 7 with ground and used wires to input-output segment. ~~The~~

That's how, using 4 universal gates (NAND) we got to know the practical applications of Boolean algebra mechanisum and came with the output of X-OR gate ~~expt~~ expressing the whole algebric equation and Truth Table as well.